Ans to Ex.2

1. (a) $\hat{y} = 22.955 - 1.832t + 0.168t^2$
   (b) DW=2.391, d_L=0.812, d_U=1.579, 4-d_L=3.188, 4-d_U= 2.421, don't reject H_0
   (c) b_1 and b_2 are both significant, R^2=0.72
   (d) Theil's U=0.033 (good model), or 0.56 from another definition of U.
   (e) Model acceptable judging from (c) & (d)
   (f) 27.591; interval= (22.54,32.63)=$27.591 \pm 1.5513(1.438)(2.262)$.
   In the prediction interval, the correction factor is given by
   $(1 + x_0'(X'X)^{-1}x_0)^{1/2}$ =(1+1.0682)$^{1/2}$=1.438. Here $x_0' = (1 \ 13 \ 13^2)$ and
   $$X = \begin{pmatrix} 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ \vdots & & \vdots \\ 1 & 12 & 12^2 \end{pmatrix}.$$

2. (a)

   r=0.942; t-statistics=7.9, reject H_0

   Linear model

   Sales=569.45+ 140.51*time
   (5.18) (7.94)

   R_{adj}^2 = 0.87, F=63, DW=1.05(I), n=10

   Exponential model

   Ln(sales)=6.56+0.11*time
   (7.82)

   R_{adj}^2 = 0.87, F=61.1, DW=1.12(I), n=10

   First difference model

   D(sales)= 51.54 + 9.21*time
   (0.42)

   R_{adj}^2 = 0.11, F=0.18, DW=1.56(N), n=9

   Logistic model

Logistic(sales)= 1.14 - 0.34*time
(-8.08)

R_{adj}^2 = 0.88, F=65.44, DW=0.95(I), n=10

Note: The answer depends on the choice of L. I don't know the choice of the author in this case. But it is a regression of Log((L/y_t)-1) on t (time).

Quadratic model

Sales=713.29 +68.59*time + 6.54*time^2

R_{adj}^2 = 0.87, F=31.37, DW=1.16(I), n=10

Cubic model

Sales=1293.06 - 445.64*time + 118.03*time^2-6.76*time^3
(-3.61) (4.63) (-4.42)

R_{adj}^2 = 0.96, F=82.95, DW=2.42(N), n=10

The following table is based on fitted values.

Model	R_{adj}^2	RMSE	MAE	MAPE	Th. U
Linear	.873(3)	143.8(4)	127.6(4)	11.03(4)	0.051(4)
Exponential	.870(5)	140.2(3)	127.8(5)	9.98(3)	0.05(3)
Logistic	0.877(2)	159.6(5)	118.5(2)	11.43(5)	0.056(5)
Quadratic	0.871(4)	135.7(2)	122.8(3)	9.78(2)	0.048(2)
Cubic	0.965(1)	65.7(1)	46.6(1)	4.36(1)	0.023(1)

Thus Cubic model is the best model according to the ranking in various criteria.

Forecasts & intervals (Cubic):

Year		low		forecast		up
91		1225.7		1679.1		2132.6
92		482.5		1265.6		2048.8
93		-672.8		601.67		1876.1

2 (b) Fitting the individual models is similar to 2(a). We have the following table similar to 2(a).

r=0.906, t-stat=8.27, reject H_0

Model	R_{adj}^2	RMSE	MAE	MAPE	Th. U
Linear	.81(2.5)	307.9(4)	235.4(3)	3.97(4)	.026(4)
Fir. Diff	0.07(5)	334.4(5)	237.3(5)	3.96(3)	0.028(5)
Logistic	0.79(4)	292.5(3)	229.7(1)	3.86(1)	0.025(3)
Quadratic	0.82(1)	289.1(2)	232.6(2)	3.9(2)	0.024(1.5)
Cubic	0.81(2.5)	285.4(1)	236.5(4)	4.03(5)	0.024(1.5)

Thus according to average rank, quadratic is the best model.

Forecasts and intervals (quadratic)

Year	low	Forecast	up
89	5954.15	6838.85	7723.56
90	5900.67	6878.75	7856.83
91	5813.53	6908.73	8003.93

3. (a) , (b), and (c)
SSE(SES)=57557, SSE(MA3)=61635, hence SES is better.

4. We know $E(S_T) = E(x_T) - \frac{1-\alpha}{\alpha} b_2$ (1)

Now $S_T^{[2]} = \alpha S_T + (1-\alpha) S_{T-1}^{[2]}$

$= \alpha S_T + (1-\alpha)\left[\alpha S_{T-1} + (1-\alpha) S_{T-2}^{[2]}\right]$

$= \dots$

$= \alpha \sum_{j=1}^{T-1} (1-\alpha)^j S_{T-j} + (1-\alpha)^T S_0^{[2]}$

Thus as T → infinity, the second term vanishes. Take expectation on both sides of above equation, and by (1)

$E(S_T^{[2]}) = \alpha \sum_{j=0}^{\infty} (1-\alpha)^j \left[E(x_{T-j}) - \frac{1-\alpha}{\alpha} b_2 \right],$

$= \alpha \sum_{j=0}^{\infty} (1-\alpha)^j [b_1 + b_2(T-j) - \frac{1-\alpha}{\alpha} b_2]$, because $E(x_T) = b_1 + b_2 T$

$= b_1 + \alpha \sum_{j=0}^{\infty} (1-\alpha)^j b_2 (T-j) - \frac{1-\alpha}{\alpha} b_2$

$= b_1 + b_2 T - \frac{1-\alpha}{\alpha} b_2 - \frac{1-\alpha}{\alpha} b_2$

$= E(S_T) - \frac{1-\alpha}{\alpha} b_2$